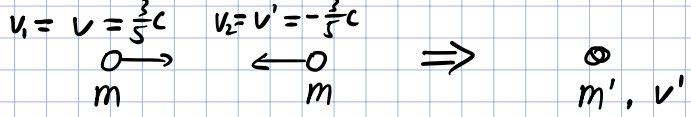
Examples 3.1




conservation of momentum:

$$\frac{mv_1}{\sqrt{1-\frac{v_1^2}{c^2}}} + \frac{mv_2}{\sqrt{1-\frac{v_2^2}{c^2}}} = 0 = \frac{m'v'}{\sqrt{1-\frac{v'^2}{c^2}}} \Rightarrow v' = 0$$

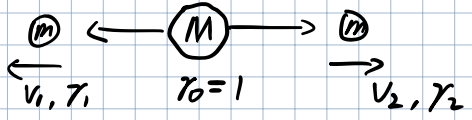
$\frac{5}{2} \cdot (\frac{3}{5})^2$

conservation of energy:

$$\frac{mc^2}{\sqrt{1-v^2/c^2}} + \frac{mc^2}{\sqrt{1-v^2/c^2}} = \frac{5}{2}mc^2 = \frac{mc^2}{\sqrt{1-0}} = m'c^2.$$

$$\Rightarrow \quad m' = \frac{5}{2}m$$

Example 3.2.




Conservation of momentum:

$$\frac{mv_1}{\sqrt{1-v_1^2/c^2}} + \frac{mv_2}{\sqrt{1-v_2^2/c^2}} = \gamma_1 mv_1 + \gamma_2 mv_2 = Mv_0 = 0.$$

$$\Rightarrow \frac{v_1}{\sqrt{1-v_1^2/c^2}} = \frac{-v_2}{\sqrt{1-(-v_2)^2/c^2}} \Leftrightarrow v_1 = -v_2 := v$$

$$\gamma_1 = \gamma_2 = \sqrt{1-\frac{v^2}{c^2}}.$$

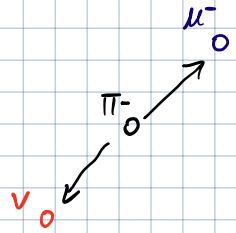
Conservation of energy:

$$\gamma_1 mc^2 + \gamma_2 mc^2 = \gamma_0 Mc^2 = Mc^2 \Rightarrow \gamma_1 + \gamma_2 = \frac{M}{m}.$$

$$\therefore \quad 2\sqrt{1-v^2/c^2} = \frac{M}{m} \Leftrightarrow v^2 = \left(1+\frac{M^2}{4m^2}\right)c^2$$

$$\begin{cases} v_1 = \sqrt{1+\frac{M^2}{4m^2}} \cdot c \\ v_2 = -v_1. \end{cases}$$

Example 3.3




conservation of momentum:

$$\frac{m\vec{v}_\nu}{\sqrt{1-v_\nu^2/c^2}} + \frac{m_\mu \vec{v}_\mu}{\sqrt{1-v_\mu^2/c^2}} = \vec{p}_{\pi^-} = 0 \Rightarrow \frac{m_\nu v_\nu}{\sqrt{1-v_\nu^2/c^2}} = \frac{m_\mu v_\mu}{\sqrt{1-v_\mu^2/c^2}}$$

conservation of energy:

$$\frac{m_\nu c^2}{\sqrt{1-v_\nu^2/c^2}} + \frac{m_\mu c^2}{\sqrt{1-v_\mu^2/c^2}} = m_\pi c^2 \Leftrightarrow \frac{m_\nu}{\sqrt{1-v_\nu^2/c^2}} + \frac{m_\mu}{\sqrt{1-v_\mu^2/c^2}} = m_\pi .$$

$$E_\nu = \left(\frac{m_\pi^2 - m_\mu^2 + m_\nu^2}{2m_\pi}\right) c^2 .$$

$$E_\mu = m_\pi^2 c^2 - E_\nu = \left(\frac{m_\pi^2 + m_\mu^2 - m_\nu^2}{2m_\pi}\right) c^2 = \frac{m_\pi c^2}{\sqrt{1-v_\pi^2/c^2}}$$

$$\Rightarrow v_\nu = \sqrt{\left[1 - \left(\frac{2m_\pi^2}{m_\pi^2 + m_\mu^2 - m_\nu^2}\right)^2\right] c^2}$$